Scheme and answer key for

QP CODE: 25901445

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEPHY207 – CURRENT ELECTRICITY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any 10 questions

Each question carries 2 marks

1. State the sign convention used while applying Kirchhoff's Voltage Law.
   A rise in voltage should be given +ve sign and a fall in voltage a –ve sign. If one go through a resistor in the same direction as the current, then the voltage is –ve and vice versa. (Any point 2 mark).

2. What are the factors that affect the resistance of a conductor?
   Factors are
   (1) Length – varies directly as the length of the conductor
   (2) Area of cross section – varies inversely as the cross section of the conductor
   (3) Nature of the material
   (4) Temperature of the conductor
   (Any 2 points- 2 marks / Give full marks for the formula $R = \rho \frac{l}{A}$).

3. What is the condition for the maximum power transfer in a circuit?
   Maximum power is transferred when the load resistance is equal to the resistance of the network as viewed from the output terminals with all the energy sources removed leaving behind their internal resistances. (Give full marks for this point).

4. Define electric current? What is its unit?
   Rate of flow of charge/ $i = \frac{dq}{dt}$ (1 mark)
   Unit-Ampere (1 mark)

5. What are the practical units of capacitance?
   Microfarad, nanofarad, picofarad, farad, etc (Any two- 2 marks)

6. Explain Faraday's law of electromagnetic induction.
   First Law- Whenever the magnetic flux liked with a circuit changes, an emf is induced. (1 mark)
   Second Law- The magnitude of the induced emf is equal to the rate of change of flux linkages. (1 mark)

7. Describe a variable capacitor.
   Capacitor whose capacitance can be varied. Eg. Radio tuning. (2 marks)

8. How does the separation between the plates affect the capacitance of a parallel plate capacitor?
   Capacitance varies inversely with the separation between the parallel plate capacitor. $C = \frac{\varepsilon_0 A}{d}$ (Statement or formula give full mark).

9. Briefly explain electrolytic dissociation.
   Process where electrolytes dissociate into cations and anions when dissolved in water/solvent.
   (Full mark any points related with this).

10. Define phase angle in RC circuit.
    In RC circuit, the phase angle is the angle that current leads the applied voltage / phase diagram (1 mark)
    $\tan\phi = -\frac{X_C}{R}$, where X_c is the capacitive reactance. (1 mark) .

11. Write expression for impedance in series LCR circuit.
    Impedance $Z = \sqrt{R^2 + (X_L - X_c)^2}$, where R-resistance, X_L- inductive reactance and X_c- capacitive reactance. (2 marks).

12. Define resonance in an LCR circuit.
    Under certain frequency, $X_L = X_c$, then circuit impedance Z=R. Under this condition, the circuit is said to be in resonance. (2 marks).
    Resonance frequency $f_0 = \frac{1}{2\pi\sqrt{LC}}$ (Expression only 1 mark, Graphical representation of resonance-2 marks).

13. Why does an emf develop when two dissimilar metals are joined and junctions are kept at different temperatures?
    This is due to Seebeck effect and the arrangement is called thermocouple. (full mark for the identification of the effect and its explanation)

14. Define Thomson effect.
    When a current flows through an unequally heated metal, there is an absorption or evolution of heat throughout in the body of the metal. (2 marks)

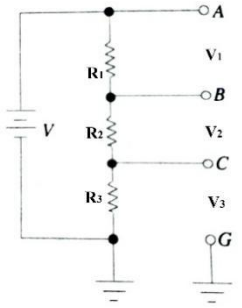
Part B

Short Essay

Answer any 6 questions

Each question carries 5 mark

15. Find the expression for V_1, V_2 and V_3 of the voltage divider circuit shown in the figure.



$V_1 = \frac{VR_1}{R_1+R_2+R_3}$, $V_2 = \frac{VR_2}{R_1+R_2+R_3}$, $V_3 = \frac{VR_3}{R_1+R_2+R_3}$

Any one-2 mark, Any two-4 mark, All three -5 mark.

16. State and explain the Superposition theorem.
According to this theorem, if there are a number of emfs acting simultaneously in any linear bilateral network, then each emf acts independently of the others.
(Statement only 3 marks, explanation with the help of a circuit – 5 marks)

17. An iron ring of 40 cm mean radius of a square iron bar cross section area 5 cm^2. A coil of 500 turns is uniformly wound over the iron ring with a wire of 0.003 cm^2 cross section area. If relative permeability of iron is 6000, calculate the self inductance of the coil.
$N = 500$; $A = 5\ cm^2 = 5 \times 10^{-4} m^2$; $l = 2\pi \times 0.4\ m$; $\mu_r = 6000$;
$\mu_0 = 4\pi \times 10 - 7$
$$L = \frac{\mu_0 \mu_r A N^2}{l} = \frac{4\pi \times 10^{-7} \times 6000 \times 5 \times 10^{-4} \times 500^2}{2\pi \times 0.4} = 0.375\ H$$

(Formula-2 mark, Substitution – 1 mark, Correct answer – 1.5 mark, Unit- 0.5 mark)

18. Derive the expression for coefficient of mutual inductance in terms of the dimensions of the coils.
Flux in first coil $\phi_1 = \frac{N_1 I_1}{l/\mu_0 \mu_r A}$ (1.5 mark)
Assuming the whole flux is linked with the other coil having N_2 turns,
$M = \frac{N_2 \phi_1}{I_1}$ (1.5 mark)

$M = \frac{N_2 N_1}{l/\mu_0 \mu_r A} = \frac{\mu_0 \mu_r A N_1 N_2}{l}$ (2 mark)

19. Show that the power consumed per cycle is zero for a circuit containing inductance only.

Instantaneous power= $V_m I_m \sin\omega t \sin\ (\omega t - \frac{\pi}{2})$ (2 mark)

$= -\frac{V_m I_m}{2} \sin 2\omega t$ (1 mark)

Power for whole cycle, $P = -\frac{V_m I_m}{2} \int_0^{2\pi} \sin 2\omega t\ dt = 0$ (2 mark)

20. What is skin effect? How does it affect transformer losses?

Skin effect is the tendency of ac to flow mainly near the surface of a conductor rather than uniformly through its cross section. As a result current density is higher near the surface and lower at the center. Effective resistance of the conductor increases. Power loss increases at high frequencies, etc (3 marks)

Skin effect mainly increases copper losses and heating in transformer windings especially at high frequencies. Also it increases eddy current loss. (2 marks)

21. What is the time constant in LR circuit? Illustrate the variation of current with time during the decay process.

The time constant of an LR circuit is defined as the time taken by the current to rise from zero to 0.632 or 63.2 % of the maximum steady value. $\tau = \frac{L}{R}$ (2 mark)

Derivation of $I = I_0(1 - e^{-Rt/L})$ (2 mark)

Graphical representation of I vs t (1 mark)

22. Obtain equation for capacitor voltage during charging and plot graph.

Derivation of the equation $V_c = V_0(1 - e^{-\frac{t}{RC}})$ (3 marks)

Graph (2 marks)

